

PlantCo Quantity Performance Analysis

Power BI Dashboard — Dynamic YTD vs PYTD Performance Report

Company	PlantCo (Fictional Plant Products Company)
Role	Data Analyst / Power BI Developer
Tool	Microsoft Power BI Desktop
Data Source	PlantCo_data.xlsx (Excel Workbook)
Period	January 2022 – April 2024
Report Type	Dynamic Dashboard with Switch Measures (YTD vs PYTD)

Overview

PlantCo is a fictional plant products company operating across multiple international markets. This Power BI project delivers a compact, interactive performance dashboard that allows stakeholders to monitor key commercial metrics on a year-to-date (YTD) basis versus the prior year-to-date (PYTD). Users can dynamically toggle between Sales, Gross Profit, and Quantity using slicer-driven SWITCH measures, and leverage conditional formatting to instantly surface underperforming markets and high-potential accounts.

Problem Statement

PlantCo requires a clear and actionable view of its commercial performance across markets, products, and customer accounts. As the data analyst, the objective is to design and build a dynamic Power BI dashboard that answers the following business questions:

- Which countries are underperforming relative to the prior year, and by how much?
- How is monthly performance trending compared to the same period last year?
- Which customer accounts are most and least profitable, and how can they be segmented?
- What is the overall gross profit margin (GP%), and how does it vary by account?

The report must support dynamic switching between Sales, Gross Profit, and Quantity metrics via a values slicer, and use conditional formatting to highlight pain points and growth opportunities at a glance.

Dataset Description

Source file: PlantCo_data.xlsx — a single Excel workbook containing three main tables and two virtual tables built within the Power BI data model.

Table 1 — Fact_Sales (Fact Table)

Primary transactional table. Each row represents a single sales invoice line.

Column	Data Type	Description
Product_ID	Integer / Text	Foreign key linking to Dim_Product
Sales	Decimal / Currency	Revenue value of the invoice line (Sales measure)
Quantity	Integer	Number of units sold on the invoice line
COGS	Decimal / Currency	Cost of goods sold; used to derive Gross Profit
Invoice_Date	Date	Invoice date; primary field used for time intelligence
Account_ID	Integer / Text	Foreign key linking to Dim_Account

Table 2 — Dim_Account (Account Dimension)

One row per unique customer account. Account_ID is the primary key.

Column	Data Type	Description
Account_ID	Integer / Text	Unique identifier for each customer account (PK)
Account_Master_ID	Integer / Text	Master-level account grouping identifier
Country	Text	Full country name of the account
Country_Code	Text	Abbreviated country code
Latitude / Longitude	Decimal	Geographic coordinates for map-based visuals
Postal_Code	Text	Postal/ZIP code for the account location
Street_Name	Text	Street address of the account
Street_Number	Text	Street number of the account

Table 3 — Dim_Product (Plant Hierarchy Dimension)

Describes the product hierarchy for all plant products. Product_Name is the unique identifier. Supports hierarchical drill-down across five levels:

Level	Description
Product_Family	Broadest grouping
Product_Group	Sub-grouping within the family
Product_Name	Specific plant product (unique identifier)
Product_Size	Size classification of the product
Product_Type	Type classification: Indoor / Outdoor / Landscape

Virtual / Calculated Tables (created in model)

- **Dim_Date** — Dim_Date

- DAX CALENDAR() table spanning 2022-01-01 to 2024-12-31. Includes an In_Past boolean column (TRUE/FALSE) to flag dates within the completed portion of the current year, enabling accurate PYTD comparisons.
- Slicer_Values —
 - Manually entered table containing three rows (Sales, Gross Profit, Quantity) that powers the values slicer and SWITCH measures.

Methodology & Approach

The project follows a four-phase structured workflow:

Phase 1 — Data Preparation (Power Query)

- Import the three tables from PlantCo_data.xlsx via the Excel Workbook connector.
- Rename tables to a consistent naming convention: Fact_Sales, Dim_Account, Dim_Product.
- Verify and correct column data types, including converting Invoice_Date to a Date type.
- Remove duplicate rows on natural key columns (Product_Name, Account_ID).
- Clean redundant columns such as Latitude_2 and Country_2.

Phase 2 — Data Modelling

- Build a Dim_Date calendar table using the DAX CALENDAR() function spanning 2022 to 2024.
- Add an In_Past calculated column to Dim_Date to support accurate PYTD filtering.
- Create the Slicer_Values table to enable switch measure functionality.
- Establish the following table relationships:
 - Fact_Sales[Account_ID] → Dim_Account[Account_ID]
 - Fact_Sales[Invoice_Date] → Dim_Date[Date]
 - Fact_Sales[Product_ID] → Dim_Product[Product_Name]
- Organise DAX measures into display folders: Base Measures, Prior-to-Date, YTD, and Switches & Comparison.

Phase 3 — DAX Measure Development

All measures are built in a dedicated Measures table and organised into the following categories:

Measure(s)	Category	Purpose
Sales, Quantity, COGS, Gross Profit	Base Measures	Core aggregations from the fact table
GP%	Base Measures	Gross Profit ÷ Sales; key profitability KPI
PYTD_Sales, PYTD_Quantity, PYTD_Gross Profit	Prior-to-Date	SAMEPERIODLASTYEAR with In_Past filter for comparable periods
YTD_Sales, YTD_Quantity, YTD_Gross Profit	Year-to-Date	TOTALYTD using Invoice_Date to stop at the latest data point
Switch_YTD, Switch_PYTD	Switch Measures	Dynamic SWITCH() toggling between Sales, Gross Profit, and Quantity based on slicer selection

Measure(s)	Category	Purpose
YTD vs PYTD	Comparison	Difference between Switch_YTD and Switch_PYTD for variance analysis
Waterfall Title, Column Chart Title, Scatter Title, Report Title	Dynamic Titles	Conditional text measures updating chart and report titles based on user selections

Phase 4 — Report Design & Visualisation

- Custom background: Designed in PowerPoint (off-white with a header band) and imported as a canvas background image for a consistent, professional look across all pages.
- Card Visuals: Header metrics display YTD, PYTD, YTD vs PYTD, and GP% with conditional colour formatting (green for positive variance, red for negative).
- Values Slicer: Enables end-users to switch between Sales, Gross Profit, and Quantity across all visuals simultaneously.
- Year Slicer: Filters the entire report by year (2023 or 2024; 2022 excluded as it has no PYTD comparator).
- Treemap: Displays the Bottom 10 countries by YTD vs PYTD, instantly surfacing the worst-performing markets.
- Waterfall Chart: Shows month-by-month contribution to the overall YTD vs PYTD variance, with drill-down by Country, Product Type, and Product Name.
- Line & Stacked Column Chart: Plots YTD performance (columns by Product Type) against PYTD (line in red) by month, with drill-up/down to Quarter level.
- Scatter Chart with Zoom Slider: Segments all customer accounts by YTD value (x-axis) and GP% (y-axis), with average benchmark lines to identify high-value and high-profitability opportunities.
- Dynamic Titles: All chart and report titles update automatically based on the selected metric and year, maintaining context for end-users at all times.

Key Findings & Insights

Insights derived from the completed Power BI report, analysing 2024 YTD data through April:

	Canada — Largest Contributor to Gross Profit Decline Canada is the primary driver of YTD vs PYTD gross profit decline, specifically due to underperformance in Landscape products during March and April 2024.
	February 2024 — Best-Performing Month February 2024 exceeded the prior-year period. Investigating the promotions, channels, or product mix active in February could help replicate this success in subsequent months.
	Bottom-10 Countries Requiring Commercial Review Canada, Colombia, Croatia, and Germany are among the worst-performing countries by YTD vs PYTD variance. These markets are candidates for immediate commercial strategy review.
	High-Margin, Low-Volume Account Segment A segment of customer accounts displays above-average GP% but below-average sales volume. These accounts represent high-priority upsell targets with strong profitability potential.
	Account Volume Concentration

The scatter chart reveals that most accounts cluster within a consistent quantity band. A small number of outlier accounts drive significantly higher invoice volumes, informing account management prioritisation.

Prerequisites & Setup

Requirements

- Microsoft Power BI Desktop (latest version recommended)
- PlantCo_data.xlsx — source Excel workbook with Fact_Sales, Dim_Account, and Dim_Product sheets
- No additional software or licenses required beyond Power BI Desktop

Getting Started

1. Clone or download this repository to your local machine.
2. Open Power BI Desktop and load PlantCo_data.xlsx via Get Data → Excel Workbook.
3. Follow the four-phase methodology outlined above to build the data model and measures.
4. Import the custom background image (PowerPoint export) as the canvas background on all report pages.
5. Publish the report to Power BI Service for stakeholder access (optional).

File Structure

File / Folder	Description
PlantCo_data.xlsx	Source data workbook containing all three input tables
PlantCo_Report.pbix	Power BI Desktop report file with full data model and dashboard
background/	Folder containing the custom canvas background image exported from PowerPoint
README.docx	This project documentation file

Licence

This project is created for educational and portfolio purposes using a fictional dataset. PlantCo and all associated data are fictitious. No real company data is represented.